Strategic & Technical Research Supplement

Vehicle Breakdown Assistance Platform

This document outlines the strategic research, legal compliance frameworks, and technical architecture recommendations for the Vehicle Breakdown Assistance Mobile Application. It is designed to supplement the core functional requirements with market-aligned data.

1. Strategic Business Model

The Aggregator Model: Adopting an "Uber for Mechanics" approach allows for low initial capital investment. Instead of owning a fleet, the platform onboards existing independent mechanics, garages, and towing operators.

Revenue Stream:
• **Commission:** 15-20% deduction from the total service invoice.
• **Surge Pricing:** 1.25x - 1.5x multiplier during night hours (11:00 PM - 5:00 AM) or severe weather.

2. Market-Aligned Pricing Matrix

The following tiered pricing structure balances competitive rates for customers with sustainable earnings for service partners:

Vehicle Tier	Service Type	Base Service Fee	Travel Rate
Two-Wheeler	Puncture / Repair	Rs. 150	Rs. 15 / km
Two-Wheeler	Breakdown Checking	Rs. 250	Rs. 15 / km
Four-Wheeler	Breakdown Checking	Rs. 500	Rs. 20 / km
Four-Wheeler	Wheel-Lift Towing	Rs. 1,500	Rs. 40 / km
Four-Wheeler	Flatbed Towing	Rs. 2,500	Rs. 60 / km
Heavy Truck	Mechanical Repair	Rs. 1,000	Rs. 25 / km
Heavy Truck	Heavy Duty Crane	Rs. 5,000	Rs. 100 / km

3. Legal & Risk Compliance Framework

A. Mobile Fuel Delivery Regulations

Constraint: Transporting petrol in loose containers (bottles/cans) is restricted under petroleum safety laws (PESO regulations) and faces police penalties.
Strategy:
• **Diesel Only:** Restrict delivery to Diesel for trucks/generators via approved partners.
• **Petrol Workaround:** For petrol vehicles out of fuel, the app should route to a "Tow to Nearest Station" service instead of delivery.

B. Trust & Safety Protocols

• **KYC Verification:** Mandatory Aadhaar and Driving License upload for all mechanics. Accounts remain inactive until document verification.
• **Geofenced SOS:** Automated alerts if a mechanic or customer remains stationary in a hazardous location for an extended period.

4. Recommended Technical Architecture

To ensure cost-effectiveness and scalability, the following tech stack is recommended:

• **Frontend (Mobile App):** Flutter (Google) - Allows a single codebase to deploy apps for both Android and iOS simultaneously.
• **Backend Engine:** Node.js with Express framework.
• **Database:** PostgreSQL with PostGIS extension (essential for accurate "Radius Search" to find nearby mechanics).
• **Live Tracking:** WebSocket protocol for real-time location updates + Google Maps API (or Mapbox for lower cost).
• **Notifications:** Firebase Cloud Messaging (FCM) for high-priority alerts.